Multiplayer Settlement Game

Beginner player guide

A complete introduction to the current shared-start prototype, with worked first steps, diagrams, every gameplay system and a full building and hardware reference.

Edition 11 October 2026 | prototype v13 plus expansion

Find your way

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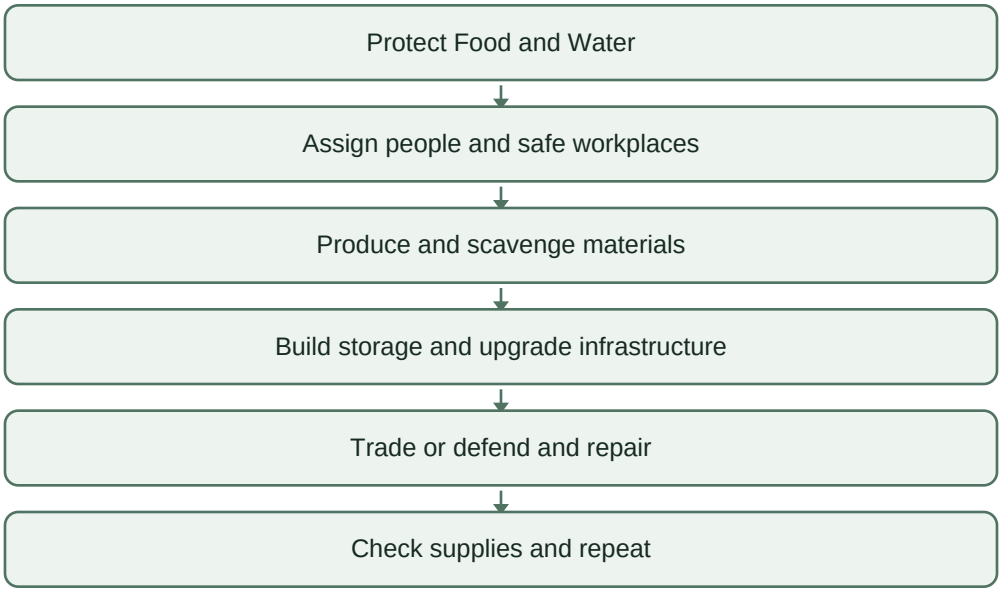
1 Start here

You lead a settlement of survivors in a shared post-apocalyptic world. Keep citizens fed, watered and safely housed, turn scavenged materials into a working economy, and decide how to trade, cooperate or fight with other settlements. This guide explains the current playable prototype from the beginning. You do not need the design documents or any programming knowledge.

This edition describes the shared-start Flask playtest using prototype v13 with its expansion, as reviewed on 11 October 2026. All current gameplay systems are enabled when the host launches this playtest. Older standalone saves can have different feature switches. Costs and rates below are the current default rules; the live interface is the authority if the host changes a future playtest configuration.

Read the first steps and resource chapters before spending your starting materials. Use the later chapters when you encounter a new feature. The final catalogue covers every building, every available upgrade and all six manufactured hardware types. In the browser use Ctrl+F to find a building or problem; the PDF is searchable and printable.

The settlement loop



Join the playtest

Open the game address supplied by the host. Create an account using a unique username of 3 to 24 letters, numbers or underscores, starting with a letter, and a password of 12 to 128 characters. Usernames are stored in lowercase. Log in, reserve a settlement name in the lobby and wait for the host to start. No settlement time runs while the lobby is waiting. All enrolled players begin together; enrollment currently closes at launch. Changing the settlement name before launch does not start the game early.

Your starting settlement

The default start gives you six level-1 citizens, a Town Hall with ten housing places, 120 Food, 120 Water, 180 Scrap, 40 Metal, 10 Electronics and 500 Credits. The Town Hall provides basic utilities and storage. Keep its Administrator assigned for its Food and Water production. The map is 40 by 40 tiles with 64 privately scavenged ruins per settlement.

Time keeps moving

The game is based on elapsed time, not turns. A completed producer works while you are away if it has workers, inputs, utilities and storage. Needs and combat also continue. Refreshing catches the save up to the current time; it does not create extra output or reroll random outcomes. Check the net supply rates before leaving the game for a long period. The host can pause the season; a paused clock earns no production and advances no need or finance timers.

2 Your first playable opening

This is a practical opening, not a requirement to build in exactly this order. It was checked using normal player actions without resource grants. Keep construction near the Town Hall and leave at least one healthy citizen free for the next job.

1. Keep the Administrator in the Town Hall. Open the map and choose empty, buildable tiles inside its Oxygen area.

2. Build a Garden for 35 Scrap. Assign a builder to finish it, then assign a healthy production worker. A completed but unstaffed Garden makes no Food.

3. Build two Rain Collectors, each costing 25 Scrap. They still need construction work, but no operating workers. Each produces 3 Water per hour at full condition.

4. Build a Salvage Depot for 55 Scrap and 20 Metal. Finish it and assign workers. At level 1 each effective worker produces 8 Scrap per hour, so two base workers produce 16.

5. Let Scrap accumulate, then add a Scrap Yard for 60 Scrap. It adds 250 shared material-storage places; this makes the first Town Hall upgrade affordable to store.

6. Use remaining workers to scavenge safe ruins for Metal and Electronics. Add an Oxygen Garden when useful ruins lie beyond current coverage. Keep checking Food and Water as survivors join.

7. Save 250 Scrap, 120 Metal and 10 Electronics for Town Hall level 2. Assign builders after paying for the upgrade. Expect its basic Food and Water production to be reduced during that work.

8. At Hall 2 plan a Warehouse, Electricity, a Recycler and more utilities before growing industry. Then collect the 450 Scrap, 250 Metal and 25 Electronics required for Hall 3.

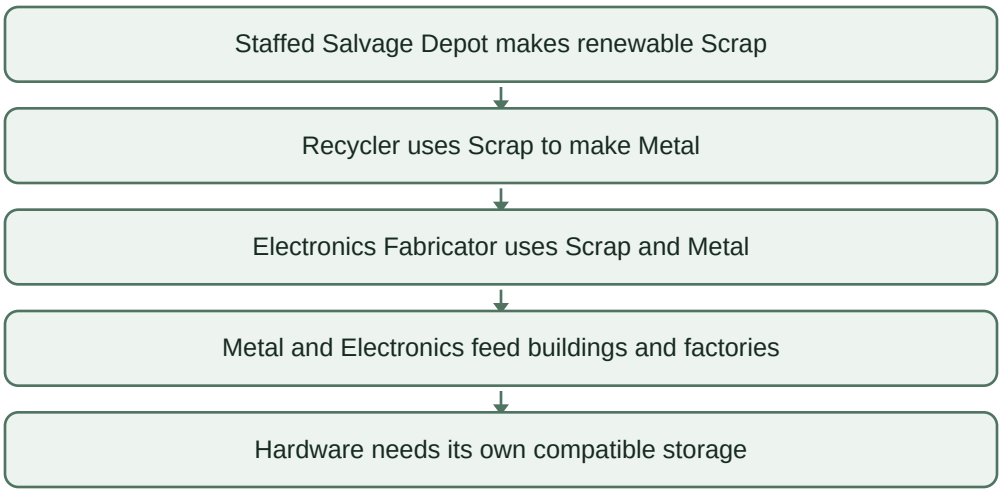
At base productivity, one Garden worker plus the Administrator provides about 8 Food per hour. Two Rain Collectors plus the Administrator provide about 8 Water per hour. Six citizens on Full rations consume 6 of each per hour. That leaves about 2 of each per hour, before injury, morale, upgrade changes or additional citizens. Watch the actual rates instead of assuming this remains enough.

The checked opening reached Hall 3 in about 66 simulated hours and built a Warehouse, Wind Turbine and Recycler along the way. Your timing will differ with placement, worker health, discoveries and choices. Do not spend the last Metal and Electronics on optional buildings before you know how you will replenish them.

3 Resources and the production chain

Resource	Where it comes from	What it is for
Food	Town Hall Administrator, Garden, Greenhouse, Hydroponics, some ruin loot, trade or raids	Hourly citizen needs. Separate Food storage.
Water	Town Hall Administrator, Rain Collector, Well/Pump, Purification Plant, some ruin loot, trade or raids	Hourly citizen needs. Separate Water storage.
Scrap	Finite ruins, staffed Salvage Depot, trade, raids, cancellation or salvage recoveries	Buildings, upgrades, repairs and converter inputs.
Metal	Ruins and Recycler conversion, trade, raids and recoveries	Advanced construction, Electronics and military hardware.
Electronics	Ruins and Electronics Fabricator conversion, trade, raids and recoveries	Advanced utilities, factories, hardware and upgrades.
Credits	Starting balance, selling on the market, direct trade, loans, investments or Credit raids	Market purchases and finance. Credits are not a physical-material storage item.
Oxygen	Town Hall, Oxygen Garden, Oxygen Generator, Industrial Oxygen Plant	Local coverage and operating capacity. It is not a stored resource.
Electricity	Town Hall, Solar Array, Wind Turbine, Generator Plant	Local coverage and operating capacity. It is not a stockpile or fuel item.

From basic materials to advanced equipment



Read stock and rates together

Stock is what you hold now. Production is what you make per hour. Consumption is what you use per hour. Net rate is production minus consumption. A positive net rate builds a buffer; a negative rate drains it. For example, 120 accessible Food with a net loss of 4 per hour lasts about 30 hours if nothing else changes. A full store can show little actual production because surplus output cannot fit.

Reservations are unavailable stock

A direct offer or market listing reserves the supplies you promised. Reserved stock still occupies storage but cannot feed citizens, pay a building cost, supply a converter or be stolen in a normal resource raid. Reserved Credits cannot be spent elsewhere. Cancel an unnecessary offer or listing to free its reservation. A large visible balance is not necessarily a large spendable balance.

Conversions spend inputs

A level-1 Recycler uses 10 Scrap per effective worker-hour to make 7 Metal. A level-1 Electronics Fabricator uses 6 Scrap and 6 Metal per effective worker-hour to make 2 Electronics. More workers increase both input use and output. Renewable Scrap is supplied before converters are processed. A converter pauses or throttles when inputs or output storage are insufficient, and citizens gain productive XP only for useful work.

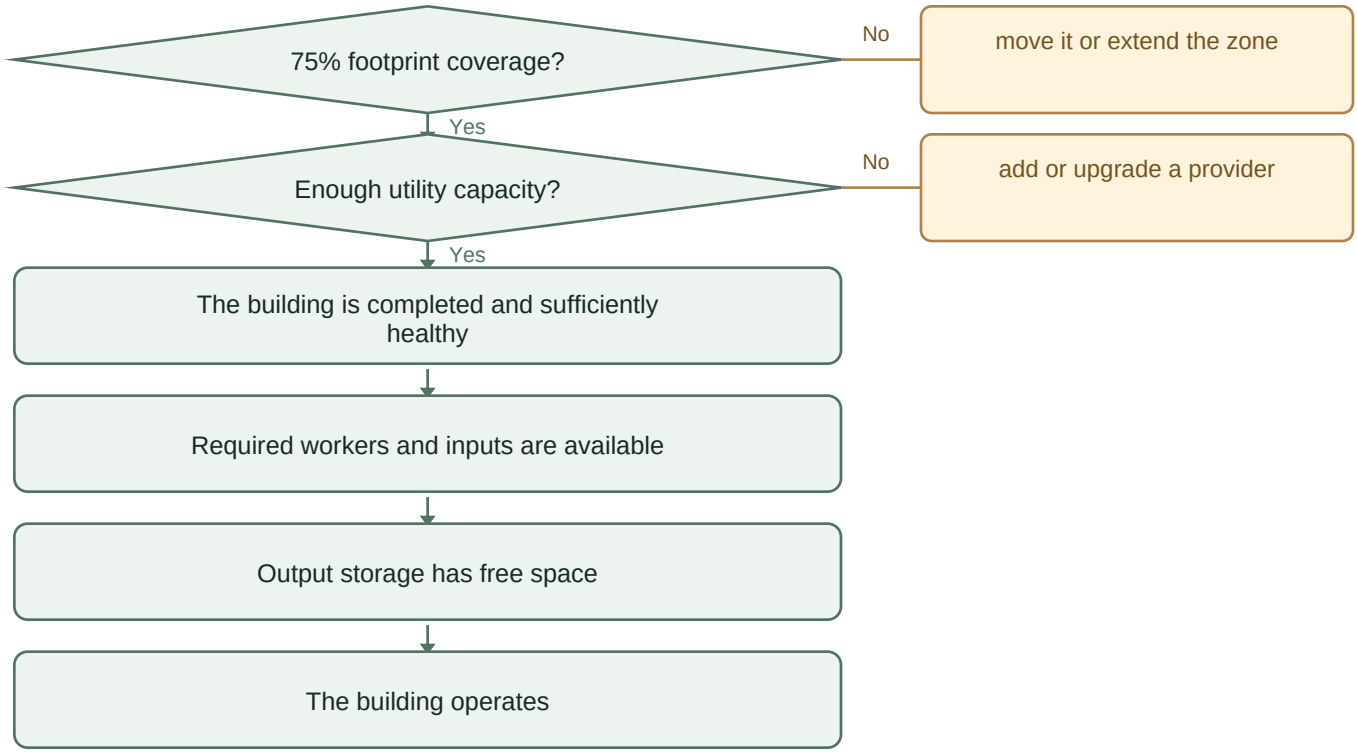
Finite loot and renewable supply

Ruins are a limited early source, especially for Metal and Electronics before the factories unlock. A Salvage Depot is the renewable Scrap source and does not consume your map ruins. Trade and raids move existing resources between settlements. A Scrap Yard is storage; it does not manufacture Scrap. An Oxygen Garden produces Oxygen, not Food.

4 Map placement and utilities

Use the map to select tiles and inspect buildings or ruins. Buildings have a rectangular footprint, often larger than one tile. The catalogue lists each size. The footprint must fit inside the map, stand on suitable land and avoid other buildings and uncleared ruins. Where offered, rotation swaps the two footprint dimensions. A red or rejected placement should be corrected before you spend supplies.

A building must pass both utility checks



Coverage is different from capacity

A provider has a radius and a limited amount of utility output. At least 75% of a required footprint must lie in coverage. Standing in the colored area is necessary but not sufficient: the provider network must also have enough capacity for demand. Several covered buildings can compete for the same limited supply. Increasing radius helps reach a distant site; increasing capacity supports more demand.

Check both Oxygen and Electricity

Many workplaces and homes need Oxygen. Advanced producers and defenses often also need Electricity. Workerless utility buildings still have construction requirements. Solar Arrays and Wind Turbines need no operating workers; an Industrial Oxygen Plant and Generator Plant do. These current generators do not have an invented fuel stock or day/night weather cycle.

Outages and priority

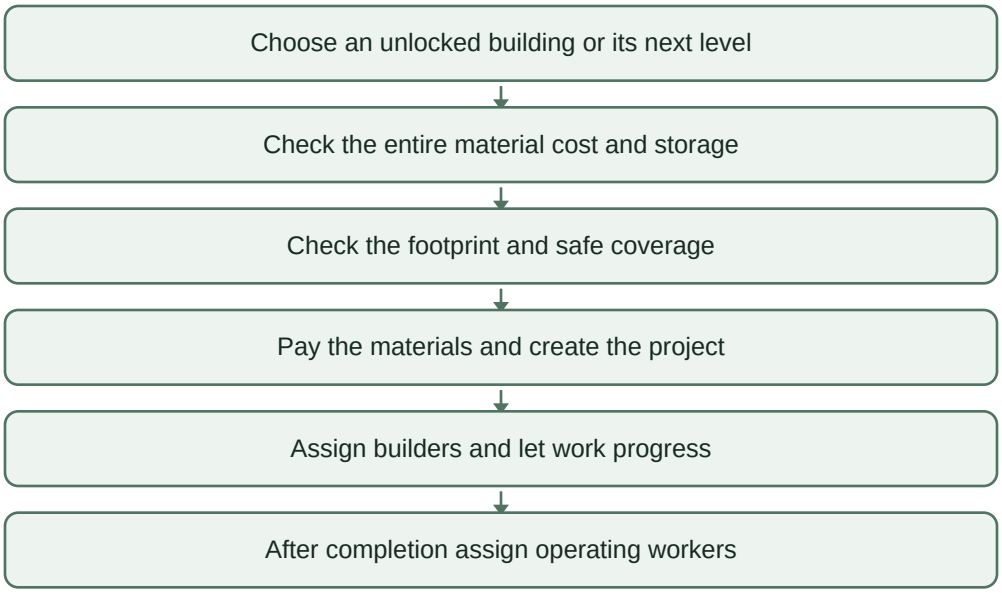
The interface explains why a building is not operating: unfinished work, damage, lack of coverage, utility capacity, staffing or inputs. Utility allocation follows building priorities and then age, so a new factory may be starved before basic services. Medical Electricity receives priority over Water in the current allocator. Repair or upgrade providers, extend coverage, or reduce demand rather than assigning additional workers to an unpowered building.

Safe work matters

An unexpected loss of workplace Oxygen evacuates workers. Deliberately unsafe assignments require confirmation and can injure citizens very quickly. At level 1 the configured Oxygen survival window is only 10 minutes, with another 0.5 minute per additional citizen level. Citizens evacuate at 25% maximum HP and deliberate exposure applies a current -10 morale modifier. This is not a practical substitute for building Oxygen infrastructure.

5 Construction and upgrades

Before starting a building or upgrade



Unlocks and levels

The Town Hall level unlocks building types. The catalogue says the Hall level required for each type and how many levels that type has. Upgrades are paid sequentially: level 1 to 2, then 2 to 3. The current rule checks the type unlock; it does not require the Town Hall level to match every upgrade level of an already unlocked type. A Garden has only one level; expand with another Garden or unlock a Greenhouse and Hydroponics. Housing types are independently buildable rather than mandatory transformation chains.

Pay once and provide builders

The full unreserved Scrap, Metal and Electronics cost is paid when the project starts. Work then progresses only with effective builders and safe conditions. Purchasing a site does not automatically assign a citizen. Several builders can help, with diminishing returns, while higher citizen level, morale and Engineering support can improve speed. The catalogue gives base work minutes, not a guaranteed real-clock completion time.

Operating staff are a separate job

Construction workers and operating workers are different assignments. When construction finishes, assign production or support workers if the building needs them. A Rain Collector has zero operating workers and produces passively once completed. A Garden requires a worker. The worker limit in the catalogue describes operating staff, not the construction crew. A worker cannot fill two simultaneous assignments.

Upgrades interrupt functions

A general building being upgraded is not an ordinary functioning producer. Housing retains usable accommodation under its special rules; the Town Hall keeps core utility zones and runs its basic Food/Water output at half its normal upgrading productivity. Plan supply buffers before taking an important producer offline. Finish or cancel an active repair before upgrading, and refresh if the interface reports a stale building level.

Cancel or demolish carefully

An unfinished construction or upgrade can be cancelled. Material refund starts at 90% and falls linearly to 25% as work approaches completion. Cancelling a new build frees its footprint; cancelling an upgrade retains the old building level. Completed buildings require demolition, which recovers 40% Scrap, 35% Metal and 25% Electronics from invested completed costs. Overflow recovery is discarded. The Town Hall cannot be demolished. Stored military assets, unfinished jobs and unsafe storage removal can prevent demolition; the interface explains what must be cleared first.

6 Storage and protecting your stock

Food, Water and physical materials have separate storage limits. Scrap, Metal and Electronics share one material pool: 200 Scrap plus 100 Metal occupies 300 places. Building costs must fit as a combined stockpile before you can pay them. The Hall-1 material store holds 250, while its first upgrade costs 380 combined materials; add storage before collecting that upgrade cost.

Need	Useful buildings	Important detail
Food space	Town Hall, Root Cellar, Cold Storage	Cold Storage needs Electricity; its outage rule differs from ordinary storage.
Water space	Town Hall, Water Tank and small producer stores	Capacity is independent of Food and materials.
Shared materials	Town Hall, Scrap Yard, Warehouse	A Warehouse is available at Hall 1 in this prototype.

Need	Useful buildings	Important detail
Drones	Their compatible hangar, bay or Workshop	Hardware slots are not resource-storage places.
Ammunition	Missile Silo and Interceptor Battery	Each type needs the correct home; launch reservations still occupy slots.
Credits	Vault	Credits need no physical store, but powered Vault allocations protect them from raids.

Condition changes capacity

Damage can reduce usable storage or hardware capacity and can cause overflow losses. Production limits output when it cannot fit. Recoveries and raid deliveries can lose excess stock rather than giving you unlimited space. Trade reservations remain protected under existing reservation checks, so capacity losses and later release are not identical to freely usable stock. Inspect available capacity before demolishing, accepting a trade or collecting salvage.

Cold Storage during an outage

Unpowered Cold Storage retains a protected portion of its Food capacity: 25% of its normal capacity in this configuration. Retained Food can be inaccessible until power returns, so visible Food does not necessarily prevent a shortage. Loss beyond usable or protected capacity can be discarded. Restore power and check accessible supplies; there is no general spoilage timer in the current game.

Vault protection

A powered Vault protects 500, 1000 or 2000 allocated Credits at levels 1, 2 or 3, at any positive HP. Loss of power removes protection while preserving the allocation. A destroyed Vault burns 25% of allocated Credits, exposes 50% to that same successful attack and retains the other 25%. Reserved trade Credits stay outside raid and Vault allocations. Inspect the Vault panel to see the actually protected amount.

7 Citizens housing health and morale

Who your citizens are

Each citizen has a persistent identity, name, sex, level, XP, HP, morale and assignment history. New citizens receive unique names. Old duplicate names are distinguished by citizen ID. Citizens can serve as Administrator, builders, production/support workers, scavengers, repair crews, patrols, guards or raiders. Captured citizens may work but cannot raid until integrated. Away citizens are unavailable for home jobs.

The Administrator

The Town Hall needs its Administrator for basic Food and Water output. You can reassign that person into ordinary work or a raid in the current version. An available citizen replaces them, chosen by level, XP and settlement seniority. If nobody is available the role remains vacant, and the Advisor warns. Releasing a worker can let a vacant Administrator role be filled. Do not assume Hall Food/Water continues at its normal rate without one.

Housing is allocated automatically

Citizens take usable safe accommodation according to housing quality and citizen priority. You do not need to drag each person into a bedroom. Damaged buildings can lose places, and Oxygen loss can make housing unusable. Housing morale bonuses apply while occupied; powered comfort can disappear in an outage. Building several named housing types is allowed: Pup Tent, Upgraded Tent, Shack, House and Apartment Block are separate catalogue entries with their own listed levels.

Work and experience

Useful production/scavenging work earns XP; construction and eligible repair work also earn XP. Waiting without effective output earns none. Citizens can reach level 10. Each extra level adds 8 maximum HP and 3% base productivity; a level increase raises the maximum without instantly healing existing damage. Low HP and poor morale reduce work and combat performance. Assign your available healthy citizens to the jobs that matter most.

Morale changes what people can do

Effective morale combines saved base morale with current housing, community, ration, exposure and debt effects, clamped to 0 to 100. Lower morale reduces productivity and may cause desertion. Ration and housing modifiers do not repeatedly accumulate each day. A new homelessness event costs 10 morale; sustained unemployment begins lowering base morale by 1 per day after 12 hours. Combat deaths and expulsion also have event effects.

Healing and deaths

Natural recovery is 2 HP per hour when Food, Water and safe housing conditions are met. Clinics and Medical Centers improve eligible care; treatment slots are automatically given to the most injured eligible citizens. No passive healing happens to drones. A citizen at zero HP dies permanently and leaves a historical record. A citizen homeless for 12 hours begins losing 1 HP per hour until safely housed.

Growth and departures at midnight UTC

Once per active UTC day, ordinary home citizens below 35 effective morale can desert, with risk rising linearly to 10% at zero morale while preserving six home citizens. One birth roll has a 12% base chance, requiring a male and female physically home, mean morale at least 40, spare safe housing, and accessible stock plus production for the next 24 hours including the newborn at current rations. Occupancy above 75% reduces the chance; full housing prevents a birth. A birth immediately creates a level-1 adult in this prototype. Reloading does not reroll outcomes.

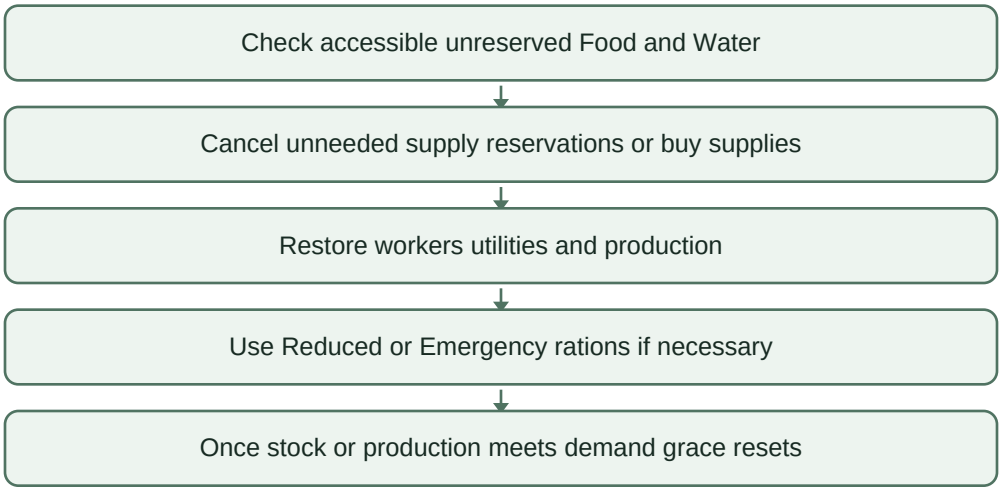
Scavenging can also discover survivors. More population means more need for supplies and housing. Midnight recovery restores at least six total living citizens when losses reduce the settlement below that floor; citizens away still count, so travel does not create free replacements. You may expel a home citizen only while leaving six home; surviving citizens lose 3 base morale. This is not a free population-management action.

8 Rations and shortage recovery

Policy	Food per person per hour	Food morale	Water per person per hour	Water morale
Full	1	0	1	0
Reduced	0.75	-3	0.8	-5
Emergency	0.5	-8	0.6	-12

Food and Water policies are selected separately. The two morale penalties add. Six home citizens on Full consume 6 Food and 6 Water per hour; six on Emergency consume 3 Food and 3.6 Water per hour. Citizens away on raids do not consume home supplies. Reduced consumption may save resources but lower morale can also reduce worker output.

Respond to a shortage



A shortage means accessible, unreserved stock is empty and production fails to cover demand. Food has a 6-hour grace period, then up to 2 HP loss per hour. Water has a 2-hour grace period, then up to 4 HP loss per hour. Partial supply scales the loss by the unmet fraction; Food and Water losses can add. Empty storage is safe if production meets demand. Restoring supply ends that shortage and resets its grace timer.

The current Emergency-ration rule has the stated morale penalty, with health damage only from actual shortages. There is no additional prolonged-Emergency health penalty while supplies meet demand. The game keeps running during supply trouble so you can recover. An administrative suspension is different: only the host can reactivate it.

9 Scavenging and support buildings

Scavenge a private ruin

Select an uncleared ruin and assign eligible scavengers with safe Oxygen access. The job reveals and delivers the ruin's saved loot progressively. Another player has their own loot state on the shared geography and does not empty your private ruins. After completion the obstacle is cleared for your settlement. Discoveries and rewards are persisted; refreshes cannot reroll them. More scavengers help, with diminishing crew speed.

Plan Oxygen before exploration

Build Oxygen Gardens toward useful uncovered ruins rather than sending citizens into dangerous work. One weak or injured worker can be slow; keep supplies positive while long jobs run. Salvage output and discoveries can exceed available storage or housing, so prepare space first. A discovered survivor can join even when housing is full and become homeless.

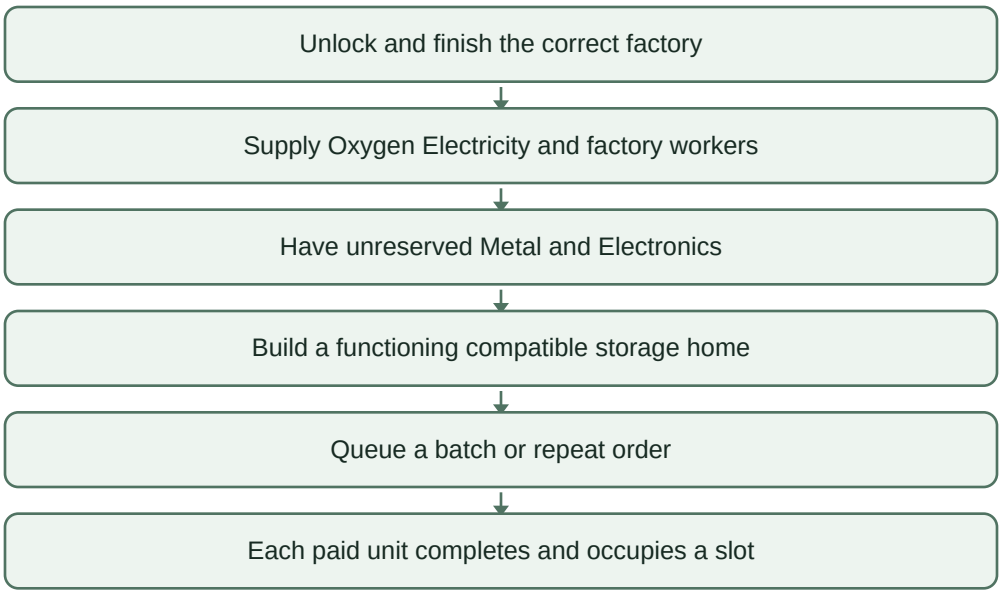
Choose support for a purpose

A functioning staffed Salvage Depot improves scavenging speed and salvage yield as well as making renewable Scrap. An Engineering Workshop improves construction/crew repairs, services damaged drones and houses Repair Drones. Clinics and Medical Centers provide treatment. A Community Center adds a current morale effect and gets a powered bonus. A Trading Post reduces your market fee. Guard Posts and patrol assignments help citizen defense.

For the same support effect, the strongest staffed, functioning building is used; duplicates do not stack the bonus. Condition and effective staffing scale support. Building another copy may add separate storage, housing or operational capacity, but should not be assumed to multiply a shared bonus. The catalogue lists the normal full-condition effects at each level.

10 Military manufacturing and hardware

Before making military hardware



The Drone Factory manufactures four drone types. The Munitions Factory manufactures Missiles and Interceptors. Factories do not supply their own complete storage needs. Build the matching home first. Manufacturing pays materials one unit at a time when that unit can start and reserves its destination slot. It pauses for insufficient workers, utilities, materials or free slots. Batch and repeat settings let later units continue automatically while conditions remain valid.

Hardware	Factory	Home	Use
Offensive Drone	Drone Factory	Offensive Drone Hangar	Raid strength, physical carrying capacity and structural damage.
Air Defense Drone	Drone Factory	Air Defense Bay	Automatic defense against offensive drones.
Ground Defense Drone	Drone Factory	Ground Defense Bay	Automatic defense when citizen raiders attack.
Repair Drone	Drone Factory	Engineering Workshop	Manual or automatic paid building repairs.
Missile	Munitions Factory	Missile Silo	Consumable random structural strike and splash.
Interceptor	Munitions Factory	Interceptor Battery	Consumable automatic missile interception.

An available drone still occupies a home slot while deployed, being serviced or committed to an order. Damaged homes can reduce capacity. A drone cannot be launched twice. Destroyed drones cannot be repaired or resurrected. Missiles and Interceptors are stored quantities rather than citizens or individually named drones. The equipment appendix gives each unit's cost, base work, HP, strength and carrying capacity.

11 Combat from declaration to return

Prepare before declaring war

Use Diplomacy or Conflict to declare war on an eligible settlement, then wait the one-hour warning. A launch normally takes one hour to arrive and consumes one of your three daily attack allowances. Those allowances reset at active UTC midnight. Beginner protection lasts 24 hours and launching your own attack ends your protection. Protected targets and active Peace/Non-aggression restrictions cannot be attacked. Keep six living citizens physically home; drone-only attacks can launch with the starting six.

Choose eligible participants

Citizen raiders must belong to you, be physically home and available, have at least 25% maximum HP and be integrated if captured. Offensive drones must be alive, available and based in functioning homes. Missiles require unreserved silo ammunition. The launch form rejects invalid combinations instead of granting a partial free attack. Defending staff and hardware are selected at arrival, so another battle or outage can change the defense before your raid reaches it.

The current attack sequence



Pick the objective

A resource raid focuses on Food, Water, Scrap, Metal or Electronics. An Infrastructure raid trades physical looting for building damage. A Population raid requires citizen raiders and focuses on capture. A Credit raid targets unprotected Credits without a physical carrying limit. Surviving offensive drones can also cause structural damage after combat. Missile strikes alone do not collect Credits or ordinary loot.

How strength and injury work

Citizen strength depends on level, HP and morale; patrols have a 15% strength bonus. Drone strength depends on type and current HP, with a living-drone strength floor of 20%. Combat adds saved random variance of up to 10%. A clean win requires the configured 10% strength margin; close contests take no ordinary physical loot. Each engaged phase injures a clean winner by 10% maximum HP, a loser by 30%, or both sides by 20% in a contest. A drone engaged in air and raid phases can take both injuries.

Defense roles are different

Air Defense Drones and Anti-Air Turrets oppose offensive drones in the air phase. Ground Defense Drones and Automated Turrets participate against living citizen raiders in their appropriate ground phase. Idle eligible home citizens, patrols and local guards can defend. Turrets defend matching nearby targets and require power. Guard Posts defend the Town Hall within their listed radius. A defense treaty is not a promise that every unit in every ally automatically fights.

Physical loot and deliveries

A successful resource fight needs survivors and meaningful remaining strength. Each surviving citizen carries 12 physical units plus 0.5 per extra citizen level; each surviving offensive drone adds 20. Resource priority starts with your chosen objective. Reserved or inaccessible stock is not ordinary loot. Your own storage can discard part of a delivery. The private report distinguishes stolen, delivered and lost-to-overflow quantities.

Away drones whose home is destroyed

Their mission proceeds normally. You receive a warning to rebuild compatible storage. Combat, earned rewards and structural effects are resolved before return slots are checked. A surviving drone relocates to a functioning compatible home with a free slot. If no slot is available it is destroyed on return, and the report records that loss. Earned rewards are not reversed. Drones still stored in a destroyed home are destroyed with it.

Missiles and explosions

Each Missile deals a 300-HP main hit to a randomly selected eligible building and 90 splash HP to nearby targets within the configured one-tile footprint distance. Each defending Interceptor attempts to stop one incoming missile, with 55%, 65% or 75% success according to battery level. Local Interceptors are used before eligible allied stock. Consumed ammunition is gone. A destroyed munitions store explodes for 180 HP per stored round with distance falloff; a store explodes only once in a chain. Stored ammunition is therefore a structural risk.

Captures and reports

Successful focused Population raids roll 25% for each eligible surviving home defender, capped at three captures. Other successful raids have 2% incidental capture for participating defenders; failed surviving raiders face 8% capture. Captures preserve identity, level, XP and HP. Captives begin at 20 morale, may work immediately and cannot raid until integration. At the next active UTC midnight, a low-morale captive faces a desertion chance rising to 50% at zero morale; surviving captives integrate and do not receive a second ordinary desertion roll that night.

Incoming countdowns hide attacker identity and composition before arrival. After resolution the two main participants get private detailed battle reports. Public news identifies the settlements and result without publishing inventories or exact loot. Contributing allies get their own contribution notice. System recalls due to protection, diplomacy, suspension or season closure return eligible assets and refund the allowance once; returned hardware still follows home availability rules.

12 Damage repairs and rebuilding

Building condition is practical

Condition is current HP divided by the completed level's maximum HP. Damage can lower output, support strength, storage, housing or defense. Most ordinary operating buildings stop at 25% condition or below; housing, Vaults and the protected Town Hall have special rules. Structural hits can also injure occupants by the building's actual HP-loss fraction of their own maximum HP. Repairing the building is not the same as healing those people.

Pay for a building repair

Select a damaged completed permanent building and start its repair project. The ordinary material charge is 50% of the level's building cost multiplied by the missing-health fraction. Assign repair workers for a base 20 HP per effective worker-hour, modified by applicable Engineering support. Tents, ruins, construction sites and upgrading buildings are not ordinary repair targets. The Town Hall also regenerates at 1 HP per hour when its rules allow; it is not instantly healed by an upgrade.

Use repair drones or service damaged drones

A Repair Drone based at an operating staffed Workshop can contribute 10 HP per hour at full drone health to a paid building repair, scaled by its HP. Several drones can share the same paid project without charging the materials again. Automatic mode chooses the lowest-condition eligible building and starts a project only when affordable; it may spend your materials. Stopping an order stops the drone contribution, not the funded repair project. Repair Drones can work beyond human Oxygen coverage.

To restore a surviving damaged drone, select a staffed functioning Engineering Workshop. It services one drone at a time at 10 HP per hour. Pay 25% of that drone's manufacture material cost multiplied by its missing-health fraction. A deployed, destroyed, full-health or foreign drone cannot be serviced. A drone being serviced cannot raid or defend, and its original slot remains reserved.

Clear destroyed buildings

Most destroyed buildings leave private salvageable ruins occupying their footprint. Assign a clearing crew to recover materials gradually, then rebuild once the tiles are cleared. The base clearing work equals the completed construction/upgrade work invested, with scavenging crew speed and applicable Depot bonuses. Base recoveries are 35% Scrap, 30% Metal and 20% Electronics of invested costs. Unfinished construction scales salvage by completed work. Tents leave no persistent ruin. Recovery overflow is discarded; repeated actions do not grant it again.

The Town Hall and rebuilding protection

Hostile damage cannot destroy your Town Hall and stops at 25% of its current maximum HP. Its basic utility zones survive. A hostile attack that leaves every other building and construction/upgrade site destroyed triggers a 24-hour rebuilding shield. An unstaffed or unpowered surviving building still prevents this shield. Later incoming attacks are recalled. Voluntary demolition cannot trigger protection. Another rebuilding shield requires recovery of three functioning non-Hall buildings after the previous shield; launching an attack ends your protection.

13 Trade market and diplomacy

Private direct trades

Choose a settlement and propose the resources or Credits you offer and what you want back. Offered supplies are reserved immediately. The recipient accepts or rejects; the proposer may cancel. Acceptance checks both settlements, unreserved requested stock and receiving capacity in one transaction. Direct trades have no fee, expire after 24 hours and allow up to ten open outgoing offers. You cannot spend the same reservation twice.

Sell and buy on the public market

List a whole quantity of Food, Water, Scrap, Metal or Electronics at a whole-Credit unit price. Unsold stock is reserved, and you can have up to twenty open listings. Buyers can partially fill a listing; buying from yourself is rejected. Listings have no automatic expiry, and cancellation is free. You must have spendable Credits and space for the purchase. A trade agreement reduces fees but does not create goods.

The seller pays the market fee from proceeds. Its base rate is 8%, reduced multiplicatively by the strongest functioning staffed Trading Post and then an active bilateral Trade Agreement's 25% reduction, with a 2% minimum. The final fee rounds upward to a whole Credit. Example: selling 10 units for 5 Credits each gives 50 gross; at 8% the fee is 4 and you receive 46. A one-Credit sale can leave zero proceeds because of rounding. Direct trade remains fee-free.

Treaty	Practical effect
Peace	Blocks attacks; closes an active war and recalls attacks between the pair. After cancellation, a six-hour hostility cooldown remains.
Non-aggression	Blocks attacks and recalls attacks between the pair. Cancellation leaves a twelve-hour hostility cooldown.
Trade Agreement	Reduces the seller market fee between the pair; does not itself prevent attacks.
Mutual Defense	Enables direct allied hardware contributions when your partner is defended. It does not itself prevent attacks.
Full Defense	Also enables current direct allied hardware contributions. Duplicate defense treaties do not stack. It does not itself prevent attacks.

Treaty	Practical effect
War	A unilateral declaration starts the one-hour warning before legal attacks; it does not override protection or a blocking treaty.

Bilateral treaties need the other player's acceptance. Proposals expire after 48 hours and can be withdrawn or rejected. Either party can cancel an active treaty. The same treaty cannot be proposed again for twelve hours after cancellation. Multiple different treaty types can coexist. Do not assume a defense or trade treaty is a non-aggression pact.

Understand allied defense

Only direct active defense partners help; support never chains through an ally's ally. Each partner offers 25% of its available eligible defensive drones and Interceptors, rounded down separately by type. Four eligible units offer one; three offer none. Defender hardware is used first, then allies are drawn evenly only as needed. Turrets stay local. Outages, deployment and servicing can remove hardware from the eligible pool. An ally's spent ammunition and damage come from that ally's actual inventory.

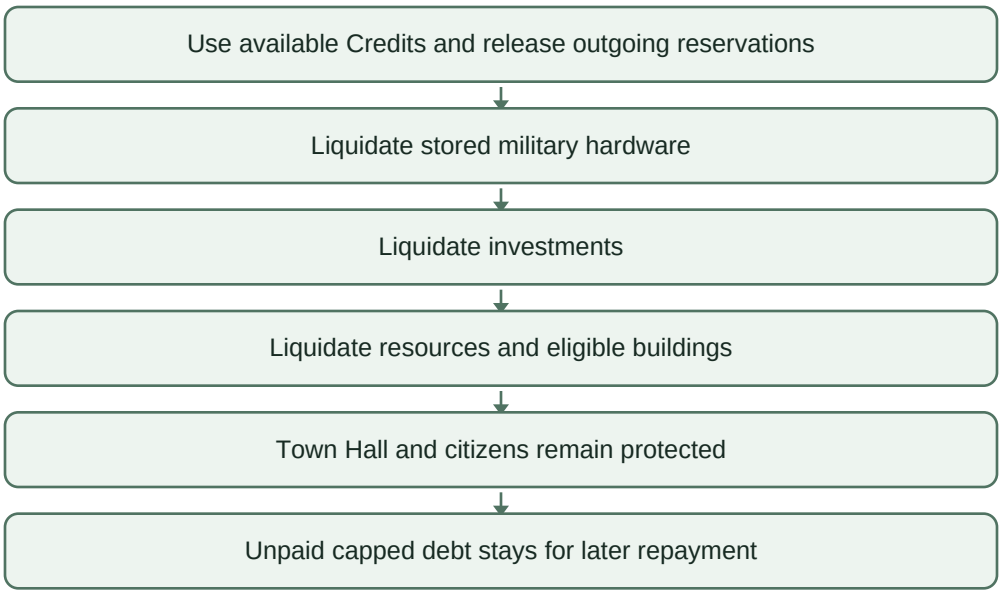
14 Loans investments and Bank risk

Borrow only with a repayment plan

The Bank offers loans with a 50-Credit minimum and total remaining principal limited to 35% of settlement value. Borrowed cash is offset by principal when collateral is calculated, so repeatedly borrowing does not create unlimited eligibility. New loan rates start at 2% per day and may rise with Bank reserve pressure; each contract fixes its rate at opening. Interest is capitalized at UTC midnight, prorated for the first interval and rounded up. Debt is capped at twice the original principal.

You may repay partly. Repayment pays accrued interest before principal. Debt also imposes a reversible current morale penalty, beginning at the configured 20% debt ratio and reaching -15. Paying debt can restore that modifier. Compare the contract's shown rate and debt with likely market income instead of treating loans as free starting Credits.

What happens when debt reaches its cap



Understand liquidation

Liquidation at the debt cap uses cash, then sells assets in the shown order at configured values. Deployed hardware cannot be sold. It can cancel outgoing reservations to make stock available. Buildings and utility loss can disrupt your economy. The Town Hall is protected and no citizens are automatically removed, but a settlement can still lose productive infrastructure. Every financial sale is recorded.

Investments are contracts

Available terms are 3, 7 or 14 days, with base simple daily returns of 0.75%, 1% or 1.25%, adjusted for Bank health when opened. Interest is not daily compounding. Whole earned Credits pay with retained fractional carry at maturity, automatically once. Early withdrawal returns principal and forfeits accrued interest. Eligibility conservatively subtracts outstanding borrowed principal from available Credits, so loans cannot be recycled into investments. Invest only a buffer you can leave unavailable.

Market fees help fund the common Bank reserve, and joined settlements contribute a configured reserve allocation at launch. The Bank panel shows reserve health; it affects new contract rates. Complete Bank-failure gameplay is disabled in the prototype. Existing payouts can push the reserve negative. Do not expect the final season banking and awards system to be complete yet.

15 Messages Advisor scoring and account tools

Navigate the settlement

Settlement/lobby returns to your main view. Panels and links expose building placement and jobs, scavenging, supplies/rations, citizen wellbeing, manufacturing, maintenance, trade, Conflict, Market, Diplomacy, Bank, Messages and Board. Read each building's pause reason and the project progress display. Completed buildings, active jobs and available workers are different lists. Help opens this guide; its first steps provide the shortest useful reading path.

Ask Ash for a status report

The settlement Advisor uses local game rules to point out supply decline, full storage, housing trouble, low morale, damage, idle people, missing Administrator, paused work, operating problems and debt. Save a report when you want a record; requests are limited to one per minute, and the ten latest saved reports are shown. This is an advisory summary, not an automatic manager or an AI that changes your settlement. It does not yet forecast every defense threat.

Use private messages and the board

Messages have an inbox, outbox, read status and separate archives. Only the sender and recipient see a message. Archiving your copy does not delete the other person's copy. Subjects allow 80 characters and bodies 1000. The season board displays the latest fifty visible posts, up to 1000 characters each. The host can moderate posts in the separate console. There are request limits of twenty private messages or ten board posts per real minute. There is no instant-chat or external push-notification system.

Check news and notifications

News publishes war, treaty and raid developments without private inventory details. Notifications include messages, trades, diplomacy, incoming/resolved attacks, completed construction/upgrades/repairs/manufacturing, survivor and citizen outcomes, and finance. A destroyed home with drones away generates a rebuild warning. Mark notifications read to track new events. Communication remains available during a gameplay pause; finished archives reject new posts and messages while retaining history.

Understand the leaderboard

Population, Military, Wealth and Development each contribute 25% of normalized Overall score relative to the season category leaders. Population rewards living citizens and level; Military counts people, surviving hardware and defenses; Wealth counts valued supplies, Credits and investments minus debt; Development values invested completed work and condition with level bonuses. Damaged construction loses value. Negative Wealth remains negative. Ties compare Development, Population, Wealth and Military, then exact ties share rank. Public totals do not expose private inventories or maps.

Account and host controls

Account settings let you change your password and log out. The first registered account does not automatically become an administrator. The host's independent console can start/pause/resume play, suspend/reactivate accounts, moderate, diagnose issues, reset passwords and make audited corrections. Players do not have those powers. Ask the host for access problems; there is no email-based recovery flow. The manual does not require access to the admin console to play.

16 Daily timing and current season boundaries

The current default season lasts 30 days. Check the displayed season clock. Times are measured from the shared launch and daily processing uses midnight UTC, which may differ from your local midnight. Captive integration/desertion is processed before finance, then ordinary citizen population checks and daily allowance resets. Combat arriving at that timestamp follows that settled midnight state. Clock order and repeated requests do not create extra births, refunds or payouts.

During the final three hours, new construction/upgrade/repair/manufacturing orders, raids and new exchanges are restricted by their controls. Already funded work may finish; cancellation and relevant communication/history controls can remain available. Check an action's message instead of assuming a grey button is a missing resource. At current closure, assets and financial state receive the implemented settlement/freeze handling; automatic next-season creation, complete archives, final awards and all-time wins remain deferred.

Pausing moves relevant elapsed-time boundaries rather than retroactively charging citizens or loans for paused time. Closing a browser is not a pause. A suspended account is frozen by the host, not by lack of Food. Future public hosting, final art, richer HUD and full season rollover are separate additions; the currently functioning economy, combat, interaction and account systems are the subject of this guide.

17 Troubleshooting and a daily checklist

Problem	Check this first
A finished building does nothing	Operating workers, utilities, condition, converter inputs and free output capacity. Check its explicit reason.
A new site makes no progress	Builders, safe Oxygen, available citizen health and the paused/season state. Buying the site is not assigning a builder.
An upgrade looks affordable but is rejected	Spendable stock after reservations, exact next level, a conflicting repair and the final window.
Citizens suffer despite visible Food	Unreserved accessible stock, Cold Storage power, production versus demand, Water, Oxygen and homelessness.
The utility area is colored but the building is offline	75% footprint coverage and remaining shared capacity; colored coverage alone is insufficient.
Military manufacture stalls	The correct staffed factory, its utilities, paid inputs, a compatible functioning home and free reserved capacity.
A raid cannot launch	War warning, blocking treaties/protection, daily allowance, participant health/integration and six living home citizens.
Drones away lost their home	Read the warning and rebuild eligible storage with free slots before return. Rewards already earned remain.

Problem	Check this first
A trade fails with enough total stock	Available stock after other reservations, the other player's availability, expiry and both receiving capacities.
No rebuilding shield after heavy damage	Any surviving non-Hall building or site still prevents it, even if unpowered or unstaffed.
Account suspended or password lost	Contact the host. Feeding citizens does not remove an administrative suspension.
An action says it has changed	Refresh and resubmit only after checking the current level, balance or state; stale requests are rejected.

1. Check accessible Food/Water and net rates before assigning new jobs or going offline.

2. Read active-job pause reasons, then restore workers, inputs, utilities or output space.

3. Review home population, spare housing, health and morale before taking in more people.

4. Check treaties, incoming countdowns, drone homes, defensive power and ammunition.

5. Check offers/listings, loan interest and contract maturity; keep a repayable Credit buffer.

6. Read new notifications and private reports, repair useful infrastructure and save an Advisor report if needed.

Small glossary

HP means health points; condition is a building's remaining-HP fraction. XP means experience. A footprint is the tiles occupied by a building. Coverage is the area reached by a utility; capacity is how much demand it can serve. An assignment is one citizen's current job. A reservation is stock or a slot already promised to another operation. A modifier changes a current value while a condition holds. A ruin is an obstacle that needs clearing. UTC is the time standard used for daily checks.

18 Complete building and upgrade reference

All 42 building types and 108 configured levels are included below. In each cost column the order is Scrap / Metal / Electronics. Level 1 is the new-building price; level 2 or 3 is the additional price of that upgrade, not a cumulative replacement price. Work means base work minutes. Staff is maximum operating staff; zero still requires builders during construction. HP is completed maximum building health.

Food, Water and conversion rates are per effective operating worker-hour for staffed producers and base building-hour for workerless producers. Hall rates require its Administrator. Capacity, slots and radii are not hourly stocks. Values are nominal healthy/full-staffed effects: condition, staffing, level, morale and storage can change actual results. Bonus percentages are reductions/increases of the relevant effect, not guaranteed percentage points of the final rate. The strongest shared support does not stack.

Town Hall

Settlement anchor, core housing/storage/utilities and Administrator Food/Water. Only one starting Hall; its seeded level 1 has no purchase charge. Footprint 3 x 3 tiles. Unlock: Hall 1. Operating requirement: no required operating utility; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	0 / 0 / 0	0	1	1000	2 Food/h; 2 Water/h; 150 Food storage; 150 Water storage; 250 material storage; 10 housing places; 4 utility radius tiles; 8 O2 capacity; 6 power capacity
2	250 / 120 / 10	120	1	1400	3 Food/h; 3 Water/h; 225 Food storage; 225 Water storage; 375 material storage; 12 housing places; 5 utility radius tiles; 10 O2 capacity; 8 power capacity
3	450 / 250 / 25	240	1	1900	4 Food/h; 4 Water/h; 325 Food storage; 325 Water storage; 550 material storage; 15 housing places; 6 utility radius tiles; 12 O2 capacity; 10 power capacity

Garden

Early staffed Food production. One level only; keep it Oxygen-covered. Footprint 2 x 2 tiles. Unlock: Hall 1. Operating requirement: Oxygen; minimum operating staff 1. Maximum level 1.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	35 / 0 / 0	15	2	150	6 Food/h; 10 Food storage; 1 O2 demand

Greenhouse

Stronger staffed Food production requiring Oxygen and Electricity. Footprint 2 x 3 tiles. Unlock: Hall 2. Operating requirement: Oxygen, Electricity; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	80 / 35 / 0	40	2	350	10 Food/h; 15 Food storage; 2 O2 demand; 2 power demand
2	70 / 55 / 5	60	3	500	17 Food/h; 25 Food storage; 3 O2 demand; 3 power demand
3	90 / 80 / 10	90	4	700	26 Food/h; 40 Food storage; 4 O2 demand; 4 power demand

Hydroponics Facility

Advanced staffed Food production requiring larger utility capacity. Footprint 3 x 3 tiles. Unlock: Hall 3. Operating requirement: Oxygen, Electricity; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	120 / 100 / 15	90	3	700	18 Food/h; 20 Food storage; 3 O2 demand; 4 power demand
2	100 / 140 / 25	120	4	950	30 Food/h; 35 Food storage; 4 O2 demand; 6 power demand
3	120 / 190 / 40	180	6	1250	45 Food/h; 55 Food storage; 6 O2 demand; 9 power demand

Rain Collector

Passive Water production after construction; no operating worker or power. Footprint 1 x 1 tiles. Unlock: Hall 1. Operating requirement: no required operating utility; minimum operating staff 0. Maximum level 1.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	25 / 0 / 0	10	0	100	3 Water/h; 8 Water storage

Well/Pump

Staffed powered Water production. Footprint 2 x 2 tiles. Unlock: Hall 2. Operating requirement: Oxygen, Electricity; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	65 / 30 / 0	35	2	300	9 Water/h; 15 Water storage; 2 O2 demand; 2 power demand
2	60 / 45 / 5	55	3	450	15 Water/h; 25 Water storage; 3 O2 demand; 3 power demand
3	75 / 70 / 10	80	4	625	23 Water/h; 40 Water storage; 4 O2 demand; 4 power demand

Purification Plant

Advanced staffed powered Water production. Footprint 3 x 3 tiles. Unlock: Hall 3. Operating requirement: Oxygen, Electricity; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	130 / 110 / 15	90	3	750	17 Water/h; 20 Water storage; 3 O2 demand; 4 power demand
2	110 / 150 / 25	120	4	1000	28 Water/h; 35 Water storage; 4 O2 demand; 6 power demand
3	130 / 210 / 40	180	6	1350	42 Water/h; 55 Water storage; 6 O2 demand; 8 power demand

Oxygen Garden

Workerless local Oxygen provider; does not grow Food. Footprint 2 x 2 tiles. Unlock: Hall 1. Operating requirement: no required operating utility; minimum operating staff 0. Maximum level 1.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	40 / 0 / 0	20	0	160	3 utility radius tiles; 5 O2 capacity

Oxygen Generator

Powered local Oxygen provider; no operating worker. Footprint 1 x 2 tiles. Unlock: Hall 2. Operating requirement: Electricity; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	60 / 30 / 5	35	0	280	5 utility radius tiles; 10 O2 capacity; 2 power demand
2	55 / 45 / 8	50	0	400	6 utility radius tiles; 14 O2 capacity; 3 power demand
3	70 / 65 / 12	75	0	560	7 utility radius tiles; 18 O2 capacity; 4 power demand

Industrial Oxygen Plant

Staffed advanced Oxygen provider requiring Electricity and its own Oxygen demand. Footprint 3 x 3 tiles. Unlock: Hall 3. Operating requirement: Oxygen, Electricity; minimum operating staff 2. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	120 / 110 / 15	90	2	750	7 utility radius tiles; 24 O2 capacity; 2 O2 demand; 5 power demand
2	100 / 150 / 25	120	3	1000	9 utility radius tiles; 34 O2 capacity; 3 O2 demand; 7 power demand
3	125 / 210 / 40	180	4	1350	11 utility radius tiles; 46 O2 capacity; 4 O2 demand; 10 power demand

Solar Array

Workerless Electricity provider. No simulated daylight or weather requirement. Footprint 2 x 2 tiles. Unlock: Hall 1. Operating requirement: no required operating utility; minimum operating staff 0. Maximum level 1.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	50 / 25 / 5	25	0	220	4 utility radius tiles; 8 power capacity

Wind Turbine

Workerless Electricity provider with larger coverage/output. Footprint 2 x 2 tiles. Unlock: Hall 2. Operating requirement: no required operating utility; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	70 / 50 / 5	45	0	400	5 utility radius tiles; 12 power capacity
2	65 / 70 / 8	65	0	560	6 utility radius tiles; 17 power capacity
3	80 / 100 / 12	95	0	760	7 utility radius tiles; 23 power capacity

Generator Plant

Staffed advanced Electricity provider; current rules have no fuel input resource. Footprint 3 x 3 tiles. Unlock: Hall 3. Operating requirement: Oxygen; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	130 / 120 / 15	90	2	800	7 utility radius tiles; 22 power capacity; 3 O2 demand
2	110 / 170 / 25	120	3	1100	9 utility radius tiles; 32 power capacity; 4 O2 demand
3	140 / 230 / 40	180	4	1450	11 utility radius tiles; 44 power capacity; 5 O2 demand

Root Cellar

Unpowered Food storage. Footprint 2 x 2 tiles. Unlock: Hall 1. Operating requirement: no required operating utility; minimum operating staff 0. Maximum level 1.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	55 / 15 / 0	25	0	240	100 Food storage

Cold Storage

Powered Food storage with special outage retention and accessibility. Footprint 2 x 3 tiles. Unlock: Hall 2. Operating requirement: Electricity; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	80 / 60 / 5	55	0	450	220 Food storage; 3 power demand
2	70 / 85 / 10	80	0	625	330 Food storage; 4 power demand
3	90 / 120 / 15	115	0	850	480 Food storage; 6 power demand

Water Tank

Separate Water storage. Footprint 1 x 2 tiles. Unlock: Hall 1. Operating requirement: no required operating utility; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	45 / 25 / 0	20	0	260	120 Water storage
2	40 / 40 / 0	35	0	360	190 Water storage
3	50 / 60 / 5	50	0	500	280 Water storage

Scrap Yard

Shared Scrap/Metal/Electronics storage. It produces no Scrap. Footprint 3 x 3 tiles. Unlock: Hall 1. Operating requirement: no required operating utility; minimum operating staff 0. Maximum level 1.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	60 / 0 / 0	20	0	300	250 material storage

Warehouse

Larger shared material storage; explicitly available at Hall 1. Footprint 2 x 3 tiles. Unlock: Hall 1. Operating requirement: no required operating utility; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	80 / 55 / 0	45	0	500	450 material storage
2	70 / 80 / 5	65	0	700	700 material storage
3	90 / 110 / 10	95	0	950	1000 material storage

Recycler

Converts Scrap to Metal per effective worker-hour. Footprint 2 x 2 tiles. Unlock: Hall 2. Operating requirement: Oxygen, Electricity; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	80 / 55 / 5	45	2	420	7 Metal/h; 10 Scrap used/h; 2 O2 demand; 3 power demand
2	70 / 80 / 8	65	3	600	12 Metal/h; 16 Scrap used/h; 3 O2 demand; 4 power demand
3	90 / 110 / 12	95	4	820	18 Metal/h; 24 Scrap used/h; 4 O2 demand; 6 power demand

Electronics Fabricator

Converts Scrap and Metal to Electronics per effective worker-hour. Footprint 2 x 3 tiles. Unlock: Hall 3. Operating requirement: Oxygen, Electricity; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	100 / 90 / 15	75	2	600	2 Electronics/h; 6 Scrap used/h; 6 Metal used/h; 2 O2 demand; 4 power demand
2	85 / 125 / 25	105	3	825	3.5 Electronics/h; 9 Scrap used/h; 9 Metal used/h; 3 O2 demand; 6 power demand
3	105 / 170 / 40	150	4	1100	5 Electronics/h; 13 Scrap used/h; 13 Metal used/h; 4 O2 demand; 8 power demand

Pup Tent

Independent basic housing with its listed morale effect. Tents are not repairable permanent structures. Footprint 1 x 1 tiles. Unlock: Hall 1. Operating requirement: Oxygen; minimum operating staff 0. Maximum level 1.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	15 / 0 / 0	5	0	60	1 housing places; 1 O2 demand; -8 housing morale

Upgraded Tent

Independent tent housing, not a mandatory conversion of an existing Pup Tent. Footprint 1 x 2 tiles. Unlock: Hall 1. Operating requirement: Oxygen; minimum operating staff 0. Maximum level 2.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	25 / 0 / 0	10	0	80	1 housing places; 1 O2 demand; -8 housing morale
2	20 / 0 / 0	15	0	100	2 housing places; 2 O2 demand; -4 housing morale

Shack

Independent basic permanent housing. Footprint 1 x 1 tiles. Unlock: Hall 1. Operating requirement: Oxygen; minimum operating staff 0. Maximum level 1.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	50 / 10 / 0	20	0	180	2 housing places; 2 O2 demand

House

Independent housing type. Its listed level-1 benefit is exactly the catalogue value; do not assume the name implies a later tier. Footprint 1 x 2 tiles. Unlock: Hall 2. Operating requirement: Oxygen; minimum operating staff 0. Maximum level 2.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	70 / 25 / 0	30	0	260	2 housing places; 2 O2 demand
2	60 / 40 / 0	45	0	360	5 housing places; 4 O2 demand; 2 housing morale

Apartment Block

Independent housing type with its own sequential upgrades. Footprint 2 x 2 tiles. Unlock: Hall 3. Operating requirement: Oxygen; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	120 / 80 / 5	60	0	500	2 housing places; 3 O2 demand
2	100 / 100 / 10	90	0	700	5 housing places; 6 O2 demand; 2 housing morale
3	120 / 140 / 20	120	0	950	12 housing places; 12 O2 demand; 5 housing morale

Engineering Workshop

Staffed construction/crew-repair support, drone servicing and Repair Drone home slots. Footprint 2 x 2 tiles. Unlock: Hall 2. Operating requirement: Oxygen, Electricity; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	80 / 60 / 5	50	2	450	2 O2 demand; 3 power demand; 10 crew work bonus %; 2 Repair Drone slots
2	70 / 85 / 10	75	3	625	3 O2 demand; 4 power demand; 18 crew work bonus %; 4 Repair Drone slots
3	90 / 120 / 15	105	4	850	4 O2 demand; 6 power demand; 28 crew work bonus %; 6 Repair Drone slots

Drone Factory

Manufactures all four drone types; requires separate compatible home slots. Footprint 3 x 3 tiles. Unlock: Hall 3. Operating requirement: Oxygen, Electricity; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	140 / 140 / 25	120	3	900	3 O2 demand; 6 power demand
2	120 / 190 / 40	165	4	1200	4 O2 demand; 8 power demand; 15 factory speed bonus %
3	150 / 260 / 65	225	6	1600	6 O2 demand; 12 power demand; 30 factory speed bonus %

Offensive Drone Hangar

Home and reserved slots for Offensive Drones. Footprint 2 x 3 tiles. Unlock: Hall 3. Operating requirement: Electricity; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	90 / 80 / 10	70	0	600	3 power demand; 6 drone slots
2	80 / 115 / 18	100	0	825	4 power demand; 10 drone slots
3	100 / 160 / 30	140	0	1100	6 power demand; 15 drone slots

Air Defense Bay

Home and slots for Air Defense Drones. Footprint 2 x 2 tiles. Unlock: Hall 3. Operating requirement: Electricity; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	80 / 70 / 10	60	0	500	3 power demand; 6 drone slots
2	70 / 100 / 15	85	0	700	4 power demand; 10 drone slots
3	90 / 140 / 25	120	0	950	6 power demand; 15 drone slots

Ground Defense Bay

Home and slots for Ground Defense Drones. Footprint 2 x 2 tiles. Unlock: Hall 3. Operating requirement: Electricity; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	85 / 75 / 8	60	0	550	3 power demand; 6 drone slots
2	75 / 105 / 15	85	0	750	4 power demand; 10 drone slots
3	95 / 145 / 25	120	0	1000	6 power demand; 15 drone slots

Munitions Factory

Manufactures Missiles and Interceptors; needs their separate storage homes. Footprint 3 x 3 tiles. Unlock: Hall 3. Operating requirement: Oxygen, Electricity; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	150 / 150 / 25	120	3	950	3 O2 demand; 6 power demand
2	125 / 200 / 40	165	4	1250	4 O2 demand; 8 power demand; 15 factory speed bonus %
3	160 / 275 / 65	225	6	1650	6 O2 demand; 12 power demand; 30 factory speed bonus %

Missile Silo

Missile stock and launch-reserved slots; stored ammunition can explode when destroyed. Footprint 2 x 2 tiles. Unlock: Hall 3. Operating requirement: Electricity; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	90 / 100 / 15	75	0	700	3 power demand; 8 munition slots
2	80 / 140 / 25	105	0	950	4 power demand; 14 munition slots
3	100 / 190 / 40	150	0	1250	6 power demand; 22 munition slots

Interceptor Battery

Interceptor stock and level-based automatic interception. Footprint 2 x 2 tiles. Unlock: Hall 3. Operating requirement: Electricity; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	90 / 90 / 15	70	0	650	3 power demand; 8 munition slots
2	80 / 130 / 25	100	0	900	4 power demand; 14 munition slots
3	100 / 180 / 35	140	0	1200	6 power demand; 22 munition slots

Clinic

Staffed medical treatment with no Electricity requirement. Footprint 2 x 2 tiles. Unlock: Hall 2. Operating requirement: Oxygen; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	70 / 35 / 0	40	2	350	2 O2 demand; 25 healing bonus %; 2 treatment slots
2	60 / 55 / 5	60	3	500	3 O2 demand; 40 healing bonus %; 3 treatment slots
3	80 / 80 / 10	90	4	700	4 O2 demand; 60 healing bonus %; 4 treatment slots

Medical Center

Stronger staffed powered medical care. Footprint 3 x 3 tiles. Unlock: Hall 3. Operating requirement: Oxygen, Electricity; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	120 / 100 / 15	90	3	750	3 O2 demand; 4 power demand; 60 healing bonus %; 4 treatment slots
2	100 / 145 / 25	120	4	1000	4 O2 demand; 6 power demand; 85 healing bonus %; 6 treatment slots
3	125 / 200 / 40	180	6	1350	6 O2 demand; 8 power demand; 120 healing bonus %; 8 treatment slots

Salvage Depot

Renewable Scrap per effective worker-hour plus strongest applicable scavenging speed/yield support. Footprint 1 x 2 tiles. Unlock: Hall 1. Operating requirement: Oxygen; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	55 / 20 / 0	25	2	260	8 Scrap/h; 2 O2 demand; 15 scavenge speed %; 5 salvage yield %
2	50 / 35 / 0	40	3	380	12 Scrap/h; 3 O2 demand; 25 scavenge speed %; 10 salvage yield %
3	65 / 50 / 5	60	4	520	18 Scrap/h; 4 O2 demand; 40 scavenge speed %; 15 salvage yield %

Trading Post

Strongest applicable staffed market-fee discount for the seller. Footprint 2 x 2 tiles. Unlock: Hall 2. Operating requirement: Oxygen; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	70 / 35 / 0	40	2	350	2 O2 demand; 10 market fee reduction %
2	60 / 55 / 5	60	3	500	3 O2 demand; 18 market fee reduction %
3	80 / 80 / 10	90	4	700	4 O2 demand; 28 market fee reduction %

Community Center

Current morale support. Its functioning bonus is strengthened by power. Footprint 2 x 2 tiles. Unlock: Hall 2. Operating requirement: Oxygen; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	75 / 40 / 0	45	2	400	2 O2 demand; 3 community morale
2	65 / 60 / 5	65	3	560	3 O2 demand; 5 community morale
3	85 / 90 / 10	95	4	760	4 O2 demand; 8 community morale

Guard Post

Local guard staffing and defense radius, used around the Town Hall in resource/citizen raids. Footprint 1 x 1 tiles. Unlock: Hall 2. Operating requirement: Oxygen; minimum operating staff 1. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	45 / 20 / 0	20	1	220	1 O2 demand; 3 defense radius tiles; 10 defense strength
2	40 / 35 / 0	35	2	320	2 O2 demand; 4 defense radius tiles; 18 defense strength
3	50 / 50 / 5	50	3	450	3 O2 demand; 5 defense radius tiles; 28 defense strength

Automated Turret

Powered local ground defense against citizen raiders. Footprint 1 x 1 tiles. Unlock: Hall 2. Operating requirement: Electricity; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	55 / 40 / 5	35	0	300	2 power demand; 4 defense radius tiles; 18 defense strength
2	50 / 60 / 8	50	0	430	3 power demand; 5 defense radius tiles; 28 defense strength
3	65 / 85 / 12	75	0	600	4 power demand; 6 defense radius tiles; 42 defense strength

Anti-Air Turret

Powered local air defense against Offensive Drones. Footprint 1 x 1 tiles. Unlock: Hall 3. Operating requirement: Electricity; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	65 / 55 / 8	50	0	380	3 power demand; 5 defense radius tiles; 22 defense strength
2	55 / 80 / 12	70	0	540	4 power demand; 6 defense radius tiles; 34 defense strength
3	70 / 110 / 20	100	0	740	5 power demand; 7 defense radius tiles; 50 defense strength

Vault

Powered allocated-Credit protection. Positive HP preserves full nominal protection; outages remove protection. Footprint 2 x 2 tiles. Unlock: Hall 2. Operating requirement: Electricity; minimum operating staff 0. Maximum level 3.

Level	S / M / E	Work min	Staff	HP	Nominal effects
1	90 / 90 / 15	60	0	700	2 power demand; 500 protected Credit capacity
2	80 / 130 / 25	85	0	950	3 power demand; 1000 protected Credit capacity
3	100 / 180 / 40	120	0	1250	4 power demand; 2000 protected Credit capacity

19 Complete hardware reference

These base unit work times are accelerated by effective factory staff, building condition and the factory's listed production bonus. Material cost is paid per unit when its work can start. The factory unlock comes from its building type; a storage home must also be unlocked and functional. No citizen XP or level system applies to hardware.

Hardware	Metal / Electronics	Work min	HP	Full strength	Physical loot
Offensive Drone	18 / 6	30	100	30	20
Air Defense Drone	16 / 7	30	100	28	none
Ground Defense Drone	14 / 5	25	120	25	none
Repair Drone	12 / 6	25	80	see role	none
Missile	20 / 8	35	quantity	see role	none
Interceptor	12 / 6	20	quantity	see role	none

Offensive Drone: strength 30 and 20 physical loot units. Air Defense Drone: strength 28. Ground Defense Drone: strength 25. Repair Drone: 10 HP/hour building repair at full health, no combat strength. Missile: 300-HP main structural hit plus splash, not a drone-combat strength. Interceptor: one interception attempt per incoming missile at the battery's configured success rate. Refer to the combat chapter for phase and target restrictions.